

Buying specificity with a dissociation: endothelial dysfunction and relaxation reserve in hard flaccid syndrome

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Eleventh in a series; see companion papers on the neural, effector, sensory and structural conjectures.

Abstract

Background. Every other conjecture in this series treats hard flaccid syndrome (HFS) as a disorder of excess contraction. This one reframes it as a disorder of deficient relaxation: chronic contraction and hypoxia impair endothelial nitric oxide production in the cavernosal sinusoids, so the tissue loses the capacity to relax rather than merely gaining the drive to contract. Its author rates it the least specific conjecture in the set, and we agree — endothelial dysfunction is invoked so widely that it explains loosely and predicts weakly.

But specificity can be bought. A hypothesis with a general mechanism becomes testable if it predicts a *dissociation* rather than a magnitude. The standard vascular pharmacology paradigm — comparing an endothelium-dependent vasodilator against an endothelium-independent one in the same tissue — does exactly that, and an established non-invasive human method exists: iontophoresis of acetylcholine and sodium nitroprusside with laser Doppler flowmetry, with published day-to-day coefficients of variation below 10% for acetylcholine and 10–20% for nitroprusside.

Support the conjecture does not claim. Spontaneously released endothelial nitric oxide has been described as tonically *suppressing* excitation of cavernosal smooth muscle, with the balance between endothelial nitric oxide and interstitial-cell prostaglandins determining the contractile state of the penis. If that is right, loss of endothelial nitric oxide raises resting tone directly rather than only reducing erectile capacity — which is the property the hypothesis needs and does not assert.

Two corrections. Prostaglandin E1 is not “downstream of nitric oxide production”; it acts through a parallel cyclic AMP pathway. The agent that genuinely tests the endothelium-independent limb is a nitric oxide donor. Separately, and in the conjecture’s favour: the normal penile Doppler findings in this population were obtained after intracavernosal vasodilator injection, which **bypasses the endothelium entirely** — so those data are uninformative about endothelial function and should not be read as evidence against this hypothesis.

The unresolved problem. Endothelial dysfunction is a disease of vascular risk factors and age, and this population is young with normal arterial inflow. And the direction of causation is undetermined: impaired endothelial function is at least as likely to follow chronic contraction as to cause it.

Keywords: hard flaccid syndrome; endothelial dysfunction; nitric oxide; relaxation reserve; acetylcholine iontophoresis; laser Doppler flowmetry; phosphodiesterase-5 inhibitors.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. Nothing in this paper is a rationale for taking any drug, supplement or nitric-oxide-directed product; the reasoning concerns what could be measured, not what should be taken. Nothing here should influence clinical management.

1. Introduction

Ten papers in this series have asked what is contracting the penis. This one asks a different question: whether the problem is partly that the penis can no longer let go.

The reframing matters clinically because it changes what a treatment failure means. Phosphodiesterase-5 inhibitors act by prolonging a cyclic guanosine monophosphate signal that must first be generated [A] [10]. A drug that amplifies an existing signal cannot work well if the signal is not being produced. If cavernosal endothelial function is impaired in

HFS, then the mixed and often disappointing response to phosphodiesterase-5 inhibitors reported in this condition [B] [1, 2] would follow without requiring patients to differ constitutionally, and without requiring fibrosis [4, 9].

The conjecture’s author describes it as the least specific in the set: endothelial dysfunction is invoked in many conditions and explains a great deal loosely without predicting much sharply, and its presence in HFS is entirely inferred [4]. That assessment is fair. This paper argues that the low specificity is a fixable property of how the hypothesis is stated rather than an intrinsic feature of the mechanism, that a dissociation design buys the specificity back, that the validated human instrument for that design already exists, and that the conjecture is currently omitting the one piece of penile-specific physiology that would most support it.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *endothelial dysfunction* *erectile dysfunction*, *flow-mediated dilation penile*, *acetylcholine sodium nitroprusside iontophoresis laser Doppler*, *endothelial nitric oxide cavernosal tone*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for validated non-invasive human measures of endothelium-dependent and endothelium-independent vasodilatation, and Section 6 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, or a physiological principle established across independent laboratories.
[B]	Single human study, human tissue study, congress abstract, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Two sources of nitric oxide, and what each is for

Erection depends on nitric oxide raising cyclic guanosine monophosphate in cavernosal smooth muscle, producing relaxation, sinusoidal expansion and subtunical venous compression [A] [10, 11]. Two sources contribute: neuronal nitric oxide from nitrergic nerves, which initiates erection, and endothelial nitric oxide from the cavernosal sinusoidal lining, conventionally regarded as contributing more to the maintenance of erection than to its initiation [A] [VERIFY] [10].

The conjecture proposes that the second is impaired. As stated, this predicts reduced erectile capacity. What it does not obviously predict is elevated *resting* tone, which is what hard flaccid syndrome is about — and that gap is the hypothesis’s central weakness as usually formulated [D].

3.2 The support the conjecture omits

There is a penile-specific finding that closes that gap, and the conjecture does not cite it.

In corpus cavernosum, both spontaneous activity of cavernosal smooth muscle and neurally released noradrenaline contract the muscle; spontaneously produced prostaglandins from interstitial cells reinforce that excitation; and **spontaneously released nitric oxide from the endothelium suppresses excitation**. The balance between spontaneously released prostaglandins and nitric oxide is described as regulating cavernosal smooth muscle tone and determining the contractile state of the penis [C] [12, 8].

If endothelial nitric oxide tonically restrains cavernosal excitation, then its loss raises resting tone *directly*, not merely by reducing the reserve available for erection [D]. That converts the hypothesis from an account of why erections fail into an account of why the flaccid penis is firm — which is what it needs to be. We regard this as the single most useful addition available to the conjecture, and note that it also links it mechanistically to two companions: the same balance is

the subject of the interstitial-cell conjecture [8], and RhoA/Rho-kinase has been reported to suppress endothelial nitric oxide synthase in the penis [C] [VERIFY] [13], which would make the calcium-sensitisation and endothelial hypotheses two ends of one pathway rather than competitors [6].

3.3 Endothelial dysfunction and erectile dysfunction in humans

The clinical association is well established. Erectile dysfunction and cardiovascular disease share risk factors, and erectile dysfunction is regarded as a marker and independent predictor of future cardiovascular events [A] [VERIFY] [14]. A systematic review and meta-analysis has reported associations between erectile dysfunction and vascular and endothelial function measures [A] [VERIFY] [14].

Relevant to the objection that this population is too young and too healthy: endothelial cell activation has been reported in men with erectile dysfunction *without* cardiovascular risk factors and without overt vascular damage [B] [VERIFY] [15]. That does not establish causation, and it may reflect erectile dysfunction causing endothelial change rather than the reverse, but it weakens the claim that endothelial abnormality requires a conventional vascular risk profile.

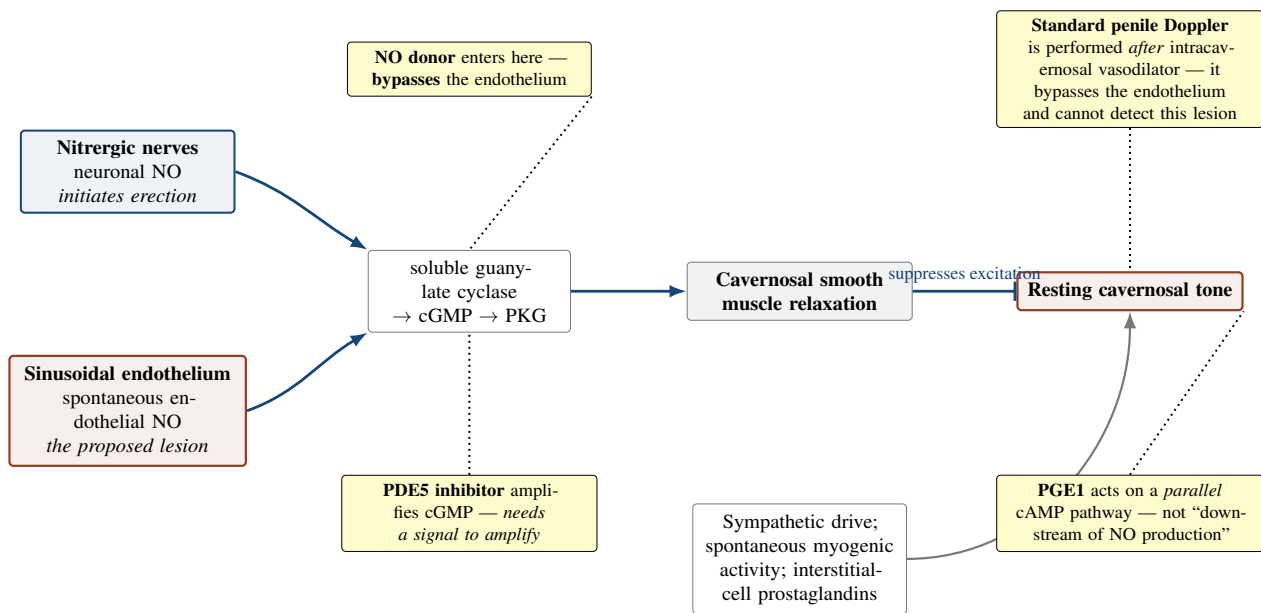


Figure 1. Where each agent acts relative to the proposed endothelial lesion. The hypothesis places the lesion at the sinusoidal endothelium. Note three consequences developed in the text. A phosphodiesterase-5 inhibitor amplifies a signal that must first be produced, so it fails when the endothelial source is impaired. A nitric oxide donor enters the cascade downstream of the endothelium and therefore tests smooth muscle responsiveness independently of endothelial function. Prostaglandin E1 acts through a parallel cyclic AMP pathway and is not, as the conjecture states, “downstream of nitric oxide production”. Finally, the normal penile Doppler findings reported in this population were obtained after intracavernosal vasodilator injection — a manoeuvre that bypasses the endothelium — so they carry no information about the lesion proposed here. The inhibitory arrow from relaxation to resting tone reflects the reported tonic suppression of cavernosal excitation by spontaneously released endothelial nitric oxide [12]; the diagram is schematic and the underlying data are from animal preparations.

4. Two corrections

4.1 Prostaglandin E1 is not a test of the nitric oxide pathway

The conjecture predicts that patients failing phosphodiesterase-5 inhibitors should include a group who respond to agents acting *downstream of nitric oxide production*, and identifies this as the pharmacological logic behind intracavernosal prostaglandin agents [4].

Prostaglandin E1 does not act downstream of nitric oxide production. It binds prostaglandin E receptors, activates adenylate cyclase and raises cyclic AMP — a parallel second-messenger pathway, not a downstream segment of the nitric oxide/cyclic GMP cascade [A] [10]. A response to prostaglandin E1 in a patient failing phosphodiesterase-5 inhibitors is therefore compatible with endothelial dysfunction but equally compatible with anything else that impairs cyclic GMP signalling, and with simple pharmacokinetic differences between an oral and an intracavernosal agent [D].

The agent that genuinely occupies the position the conjecture describes is a nitric oxide donor — sodium nitroprusside or a nitrate — which enters the cascade at soluble guanylate cyclase and therefore probes smooth muscle responsiveness independently of endothelial nitric oxide production. That substitution converts a vague prediction into

the classical endothelium-dependent versus endothelium-independent dissociation, which is where this hypothesis's testability lies (Section 6).

4.2 The normal Doppler findings do not count against the hypothesis

Among 88 men with HFS, all had normal peak systolic velocity, two had abnormal end-diastolic velocity and none had elevated echogenicity on B-mode [B] [3, 1]. Elsewhere in this series we have treated those data as informative [9]. Here they are not, and the reason is worth stating because it is easy to miss.

Penile duplex Doppler is performed *after* intracavernosal injection of a vasodilator — typically prostaglandin E1, papaverine or a combination. Those agents act directly on smooth muscle. **The examination therefore bypasses the endothelium by design**, and a normal peak systolic velocity after intracavernosal prostaglandin E1 demonstrates that the cavernosal artery can dilate when directly stimulated, not that it can dilate in response to an endothelial signal [D].

This is a point in the conjecture's favour and it generalises: *the standard urological vascular workup is structurally incapable of detecting the abnormality this hypothesis proposes*. Any study intending to test it must use an endothelium-dependent stimulus.

5. The hypothesis

Formal statement

H11 (impaired relaxation reserve). In hard flaccid syndrome, chronic contraction and reduced perfusion impair endothelial nitric oxide production in the cavernosal sinusoids. The disorder is therefore one of deficient relaxation as well as excess contraction: resting tone rises because tonic endothelial suppression of cavernosal excitation is lost, and erectile capacity falls because the signal that phosphodiesterase-5 inhibitors amplify is not being generated.

H11-tonic (the form that explains the sign). Loss of tonic endothelial nitric oxide raises resting cavernosal tone directly [12]. Prediction: endothelium-dependent relaxation is impaired while endothelium-independent relaxation is preserved.

H11-reserve (the weaker form). Endothelial function is impaired but this affects erectile capacity rather than resting tone. Prediction: the dissociation is present but uncorrelated with the severity of the hard flaccid state — a real finding with no explanatory value for the defining sign.

H11-consequence. Endothelial dysfunction is present but downstream: a marker of chronic contraction and reduced perfusion rather than a cause. Prediction: severity tracks symptom duration rather than symptom severity, and it does not precede the condition.

H11-null. Endothelial function in men with HFS is normal.

Auxiliary assumptions:

- **A1 (the tonic role is real in humans).** That spontaneously released endothelial nitric oxide tonically suppresses cavernosal excitation is reported from animal preparations [12]; the magnitude of this restraint in human tissue has not been quantified [DATA NEEDED].
- **A2 (the insult is sufficient).** Chronic contraction in a penis with normal arterial inflow produces enough hypoxia or shear-stress change to impair endothelial function. Untested, and note that this assumption is inherited from the hypoxia conjecture, whose critical link is itself contested [9].
- **A3 (penile endothelium can be inferred from cutaneous or systemic endothelium).** Required by every non-invasive test proposed below. Regional endothelial function is not uniform, and this assumption is a real limitation rather than a formality [DATA NEEDED].

6. Buying specificity: the dissociation design

The conjecture's stated weakness is that endothelial dysfunction explains loosely and predicts weakly. That is true of the hypothesis *as a magnitude claim* — “endothelial function is worse” is compatible with almost any chronic condition. It is not true of the hypothesis *as a dissociation claim*. Endothelial dysfunction has a specific signature: impaired response to an endothelium-dependent stimulus with preserved response to an endothelium-independent one. That pattern is not produced by smooth muscle disease, by fibrosis, or by excess neural drive [D].

6.1 The instrument exists and its reproducibility is published

Skin microvascular endothelial function can be assessed non-invasively by laser Doppler measurement of blood flow during transdermal iontophoresis of acetylcholine (endothelium-dependent) and sodium nitroprusside (endothelium-independent) [A] [VERIFY] [17, 18]. Reproducibility has been characterised: in one study of 16 young healthy non-

smoking men, iontophoresis of acetylcholine and of sodium nitroprusside each produced a seven- to eightfold increase in dermal blood flow, with mean day-to-day coefficients of variation on the same site below 10% for acetylcholine and between 10 and 20% for sodium nitroprusside — and, importantly, **day-to-day variation was significantly smaller than site-to-site variation** [B] [VERIFY] [16].

Three methodological cautions must be designed in rather than discovered:

- **Site-to-site variation exceeds day-to-day variation** [16], so a within-subject comparison across sites is weaker than a between-visit comparison at one site, and any patient–control comparison must standardise the site precisely.
- **Iontophoresis produces non-specific vasodilatation** from the current itself, which protocol design must minimise [B] [VERIFY] [19].
- **The acetylcholine response in human skin is not purely nitric-oxide-mediated**; prostaglandins contribute [B] [VERIFY] [20]. This is a substantive limitation: the test measures endothelium-dependent vasodilatation, not endothelial nitric oxide specifically.

Reproducibility of these laser techniques remains debated and expression of results is not standardised [B] [VERIFY] [21].

6.2 Primary test

Prediction P1. In men with HFS, the vasodilatory response to iontophoretic acetylcholine applied to penile skin is reduced relative to age-matched controls, while the response to iontophoretic sodium nitroprusside at the same site is preserved.

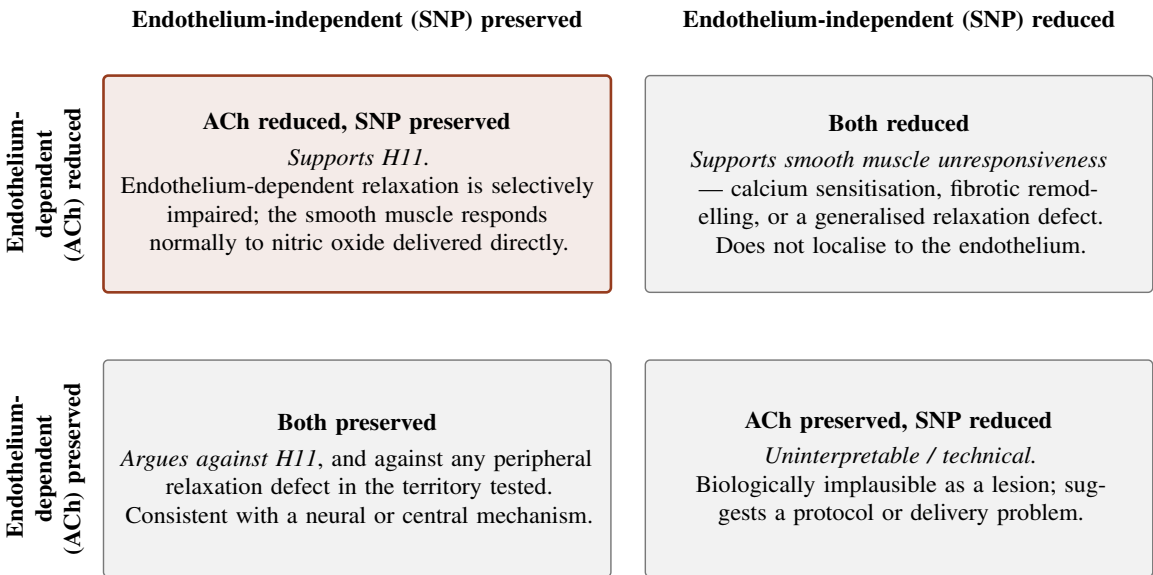


Figure 2. The dissociation grid, and why it buys specificity. A magnitude claim — “endothelial function is worse” — is compatible with almost any chronic condition. A dissociation claim is not. Only the upper-left quadrant is predicted by an endothelial lesion; the upper-right indicates a smooth muscle relaxation defect and localises elsewhere; the lower-left argues against any peripheral relaxation abnormality in the territory tested. Note the caveat in Section 7.1: acetylcholine-induced cutaneous vasodilatation is not purely nitric-oxide-mediated, so the upper-left quadrant demonstrates impaired endothelium-dependent vasodilatation rather than impaired endothelial nitric oxide specifically.

Why penile skin. It lies within the injured territory, it is accessible, and the identical instrument has been proposed in two companion papers for entirely different purposes — noradrenaline provocation to test sympathetic–sensory coupling [5] and adrenergic agonist challenge to test denervation supersensitivity [7]. **A single laser Doppler session with four iontophoresis channels could test three conjectures at once**, which is worth noting in a field with no research infrastructure. Acceptability of genital iontophoresis in this population is unknown and should be assessed first [DATA NEEDED].

6.3 Secondary tests

Prediction P2. Systemic endothelial function, measured by brachial artery flow-mediated dilation or peripheral arterial tonometry, is normal in men with HFS — indicating that any endothelial abnormality is local rather than systemic.

This prediction is as informative as P1 and cuts the other way: if systemic endothelial function is also impaired, HFS would be reclassified as a manifestation of a generalised vascular process in young men, which would be a substantial finding. We note that brachial flow-mediated dilation has high operator dependency and suboptimal repeatability, with coefficients of variation reported above 10% [B] [22], and would need a well-practised laboratory.

Prediction P3. Penile endothelial function, measured by a reactive hyperaemia protocol with venous occlusion plethysmography of the penis, is impaired relative to controls. Such a method has been described and applied in erectile dysfunction [B] [VERIFY] [23], and is more directly relevant than either cutaneous or brachial measures, but we could not establish its current availability or validation status.

Prediction P4. Among men with HFS failing phosphodiesterase-5 inhibitors, a subgroup responds to a nitric oxide donor — not, as the conjecture states, to prostaglandin E1. We do not recommend this as a study: nitrate administration with phosphodiesterase-5 inhibitors is contraindicated, intracavernosal nitroprusside is not an established agent, and the design risks are disproportionate to the information gained when the non-invasive dissociation test is available.

Table 2. Predictions of H11 compared with the principal alternatives.

Observation	H11 (endothelial)	Calcium sensitization / fibrosis	Neural drive	Discriminating value
ACH response reduced, SNP preserved	Predicted	Not predicted	Not predicted	High
Both ACH and SNP responses reduced	Not predicted	Predicted	Not predicted	High
Normal peak systolic velocity after intracavernosal vasodilator	Predicted (test bypasses endothelium)	Predicted	Predicted	None — the test cannot discriminate
Mixed or poor response to PDE5 inhibitors	Predicted	Predicted	Not specified	Low
Systemic endothelial function normal	Predicted (local lesion)	Predicted	Predicted	Low, but a positive finding would be important
Severity tracks duration rather than symptom severity	Predicted under H11-consequence	Predicted	Not predicted	Moderate

7. Alternative explanations

1. **Reverse causation.** Sustained contraction with reduced sinusoidal perfusion is an unfavourable environment for endothelial function. If endothelial impairment is found, it is at least as likely to be a consequence of the hard flaccid state as a cause of it. **Nothing in the proposed tests distinguishes these**, and a cross-sectional study cannot; only a prospective cohort followed from injury could, which does not exist.
2. **The population is wrong for the mechanism.** Endothelial dysfunction is characteristically a disorder of age and vascular risk factors. This population has a mean age of 27.4 years [2] and normal arterial inflow [3]. The counter-evidence — endothelial activation reported in men with erectile dysfunction lacking risk factors [15] — weakens but does not remove this objection.
3. **Smooth muscle rather than endothelium.** Calcium sensitisation [6] or fibrotic remodelling [9] would impair relaxation without any endothelial lesion, and would produce the upper-right rather than upper-left quadrant of Figure 2.
4. **The mixed PDE5 inhibitor response has simpler explanations.** Variable adherence, dose, timing, expectation, and the absence of any validated outcome measure in this condition. A pharmacological pattern reported from uncontrolled use [E] [4] is weak evidence for any mechanism.
5. **The premise may be false.** Cavernosal tone has never been objectively shown to be elevated in HFS, and cavernosal endothelial function has never been measured in it.

8. Implications

No clinical implications are drawn. Three research implications follow.

First, state the hypothesis as a dissociation. “Endothelial function is impaired” is not testable in a way that discriminates. “Endothelium-dependent relaxation is impaired while endothelium-independent relaxation is preserved” is.

Second, stop citing normal penile Doppler as evidence about relaxation capacity. The examination is performed after a direct-acting smooth muscle vasodilator and cannot detect an endothelial lesion.

Third, combine the protocols. Three conjectures in this series propose testing penile skin with a laser Doppler and an iontophoresis electrode using different agonists. One session, four channels, one control group.

9. Limitations

1. **No evidence in HFS.** Endothelial function has never been measured in this condition by any method.
2. **Low intrinsic specificity.** Even with the dissociation design, endothelial dysfunction is a general finding, and its presence would not establish that it causes the syndrome.
3. **Causal direction is not addressable** by any cross-sectional design, and no prospective cohort exists.
4. **The tonic-suppression finding is from animal preparations** and its magnitude in human tissue is unquantified.
5. **The surrogate problem.** Cutaneous and brachial measures are not penile cavernosal endothelium, and regional endothelial function is not uniform.
6. **The instrument is imperfect.** Acetylcholine-induced cutaneous vasodilatation is partly prostaglandin-mediated; iontophoresis produces non-specific vasodilatation; reproducibility of these laser methods is debated and results expression is unstandardised.
7. **Acceptability is unknown.** Genital iontophoresis has not to our knowledge been performed and its tolerability in this population is unestablished.
8. **The population is atypical** for the mechanism proposed: young, without vascular risk factors, with normal arterial inflow.
9. **Several citations are flagged unverified**, particularly the methodological reproducibility literature, which was traced through reviews.
10. **Confirmation risk.** This paper takes a conjecture its own author rates lowest and argues that it is more testable than supposed. That is a direction in which motivated reasoning runs, and the reader should note that increased testability is not increased plausibility.

10. Refutation criteria

- **Refuted.** Endothelium-dependent and endothelium-independent vasodilatation in penile skin are both normal in men with HFS relative to age-matched controls.
- **Substantially refuted.** Both responses are reduced — indicating a smooth muscle relaxation defect rather than an endothelial one, and localising the lesion elsewhere.
- **Demoted to H11-reserve.** The dissociation is present but uncorrelated with the severity of the hard flaccid state.
- **Demoted to H11-consequence.** The dissociation is present and tracks symptom duration rather than severity, or is absent in early disease.
- **Reclassified.** Systemic endothelial function is also impaired, indicating a generalised vascular process rather than a local penile one.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

11. Conclusion

This is the conjecture its own author likes least, and the reason given is the right one: endothelial dysfunction is invoked so widely that asserting it explains almost nothing. But that is a property of how the claim is stated rather than of the mechanism. Stated as a magnitude — endothelial function is worse — it is nearly unfalsifiable. Stated as a dissociation — endothelium-dependent relaxation impaired, endothelium-independent relaxation preserved — it makes a prediction that fibrosis does not make, that calcium sensitisation does not make, and that a neural account does not make. The instrument for that comparison is established, non-invasive, has published reproducibility, and could be applied to penile skin in the same session as two other conjectures in this series.

The conjecture also undersells itself in one specific way. As written it explains why erections fail, which is not what the condition is named for. The finding it needs and does not cite is that spontaneously released endothelial nitric oxide tonically suppresses excitation of cavernosal smooth muscle, with the balance of endothelial nitric oxide against interstitial-cell prostaglandins determining the contractile state of the penis. If that holds in humans, losing endothelial nitric oxide raises the resting tone directly — and the hypothesis becomes an account of the sign rather than of a complication.

Two things remain unresolved and both are serious. The population is young, without vascular risk factors and with normal arterial inflow, which is not where endothelial dysfunction is usually found. And no cross-sectional study can tell whether impaired endothelial function is the cause of a chronically contracted penis or its consequence — a distinction that matters enormously and that only a prospective cohort followed from the initiating injury could resolve. Neither objection prevents the measurement being made. Both should be printed on the discussion section of whatever paper reports it.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The methodological reproducibility figures quoted in Section 7 were traced through reviews and must be confirmed against the primary reports before a protocol is built on them.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [4]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Nothing in it supports the use of nitric oxide donors, nitrates, supplements marketed as nitric-oxide-boosting, or any other agent for hard flaccid syndrome.** Nitrates are contraindicated with phosphodiesterase-5 inhibitors and can cause life-threatening hypotension. Iontophoresis and intracavernosal injection are research or clinical procedures appropriate only under qualified supervision.

References

- [1] Pang KH, Feng J, Zhang Y. Hard-flaccid syndrome: a systematic review of aetiopathophysiology, clinical presentation and management. *Int J Impot Res.* 2026;38(5):393–403 (online 7 July 2025). doi:10.1038/s41443-025-01118-2. *Systematic review; 8 included studies, high risk of bias.*
- [2] Niedenfuehr J, Stevens DM. Hard flaccid syndrome symptoms, comorbidities, and self-reported efficacy and satisfaction of treatments: a cross-sectional survey. *Int J Impot Res.* 2025;37(6):458–464. doi:10.1038/s41443-024-00853-2. *Online convenience-sample survey, n = 143; mean age 27.4 years.*
- [3] Sullivan J, Salter C, Flores J, Guhring P, Parker M, Mulhall J. MP21-20 Exploring erectile hemodynamics in men complaining of hard-flaccid syndrome (HFS). *J Urol.* 2022;207(Suppl 5). doi:10.1097/JU.0000000000002554.20. *Congress abstract, n = 88; note that penile Doppler is performed after intracavernosal vasodilator and bypasses the endothelium.*
- [4] Jesterhfs. Fourteen conjectures on the mechanism [Internet]. *On Hard Flaccid Syndrome*; 8 August 2026. <https://jesterhfs.github.io/article-conjectures.html>. *Grey literature; not peer reviewed. Source of the conjecture examined here.*
- [5] Jesterhfs, Claude Opus 5. The intraganglionic loop. Preprint, 2026. *Companion paper; not peer reviewed.*
- [6] Jesterhfs, Claude Opus 5. Autonomy without a signal: RhoA/Rho-kinase calcium sensitization. Preprint, 2026. *Companion paper; not peer reviewed.*
- [7] Jesterhfs, Claude Opus 5. An amplifier without a signal: denervation supersensitivity. Preprint, 2026. *Companion paper; not peer reviewed.*
- [8] Jesterhfs, Claude Opus 5. The metronome and the amplifier: spontaneous cavernosal activity and interstitial cell prostaglandins. Preprint, 2026. *Companion paper; not peer reviewed.*
- [9] Jesterhfs, Claude Opus 5. A ratchet that has already been tested: hypoxia and fibrosis. Preprint, 2026. *Companion paper; not peer reviewed.*
- [10] Andersson KE. Mechanisms of penile erection and basis for pharmacological treatment of erectile dysfunction. *Pharmacol Rev.* 2011;63(4):811–859. doi:10.1124/pr.111.004515. *Comprehensive pharmacological review.*
- [11] Dean RC, Lue TF. Physiology of penile erection and pathophysiology of erectile dysfunction. *Urol Clin North Am.* 2005;32(4):379–395. doi:10.1016/j.ucl.2005.08.007. PMID: 16291031. *Physiological review.*
- [12] Hashitani H. Interaction between interstitial cells and smooth muscles in the lower urinary tract and penis. *J Physiol.* 2006;576(Pt 3):707–714. doi:10.1113/jphysiol.2006.116632. PMID: 16945972. *Review; source of the statement that spontaneously released*

endothelial nitric oxide suppresses excitation of cavernosal smooth muscle and that the prostaglandin–nitric oxide balance determines the contractile state of the penis. The key omitted support, Section 3.2.

- [13] Bivalacqua TJ, Champion HC, Usta MF, Celtek S, Chitale K, Webb RC, et al. RhoA/Rho-kinase suppresses endothelial nitric oxide synthase in the penis: a mechanism for diabetes-associated erectile dysfunction. *Proc Natl Acad Sci U S A*. 2004;101(24):9121–9126. *Animal experiment; cited for the mechanistic link between the calcium-sensitisation and endothelial hypotheses.* [VERIFY: CITED FROM A SECONDARY REFERENCE LIST]
- [14] Associations between erectile dysfunction and vascular parameters: a systematic review and meta-analysis. *World J Mens Health*. 2024. doi:10.5534/wjmh.230192. *Systematic review and meta-analysis.* [VERIFY: AUTHORS, VOLUME AND PAGES NOT CONFIRMED]
- [15] Bocchio M, Desideri G, Scarpelli P, Necozione S, Properzi G, Spartera C, et al. Endothelial cell activation in men with erectile dysfunction without cardiovascular risk factors and overt vascular damage. *J Urol*. 2004;171(4):1601–1604. *Human observational study.* [VERIFY: CITED FROM A SECONDARY REFERENCE LIST]
- [16] Kubli S, Waeber B, Dalle-Ave A, Feihl F. Reproducibility of laser Doppler imaging of skin blood flow as a tool to assess endothelial function. *J Cardiovasc Pharmacol*. 2000;36(5):640–648. PMID: 11065225. *Reproducibility study, n = 16 healthy young men; source of the coefficient-of-variation figures and the site-versus-day variation finding.* [VERIFY: AUTHOR LIST AND JOURNAL CONFIRMED ONLY FROM THE RECORD ABSTRACT]
- [17] Turner J, Belch JFF, Khan F. Current concepts in assessment of microvascular endothelial function using laser Doppler imaging and iontophoresis. *Trends Cardiovasc Med*. 2008. PMID: 18555183. *Methodological review.* [VERIFY: AUTHOR LIST, JOURNAL, VOLUME AND PAGES NOT CONFIRMED]
- [18] Cracowski JL, Minson CT, Salvat-Melis M, Halliwill JR. Methodological issues in the assessment of skin microvascular endothelial function in humans. *Trends Pharmacol Sci*. 2006;27(9):503–508. *Methodological review.* [VERIFY: CITED FROM A SECONDARY REFERENCE LIST]
- [19] Droog EJ, Henricson J, Nilsson GE, Sjöberg F. A protocol for iontophoresis of acetylcholine and sodium nitroprusside that minimises nonspecific vasodilatory effects. *Microvasc Res*. 2004;67(2):197–202. *Methodological study.* [VERIFY: CITED FROM A SECONDARY REFERENCE LIST]
- [20] Kellogg DL Jr, Zhao JL, Coey U, Green JV. Acetylcholine-induced vasodilation is mediated by nitric oxide and prostaglandins in human skin. *J Appl Physiol*. 2005. *Human physiological study; the basis for the caveat that the acetylcholine response is not purely nitric-oxide-mediated.* [VERIFY: VOLUME AND PAGES NOT CONFIRMED]
- [21] Roustit M, et al. Reproducibility of non-invasive assessment of skin endothelial function using laser Doppler flowmetry and laser speckle contrast imaging. *PLoS One*. 2013;8(4):e61320. doi:10.1371/journal.pone.0061320. *Reproducibility study.* [VERIFY: FULL AUTHOR LIST NOT CONFIRMED]
- [22] Marôco JL, Pinto M, Santa-Clara H, Fernhall B, Melo X. Flow-mediated slowing shows poor repeatability compared with flow-mediated dilation in non-invasive assessment of brachial artery endothelial function. *PLoS One*. 2022;17(5):e0267287. doi:10.1371/journal.pone.0267287. *Repeatability study; cited for the operator dependency and repeatability limitations of brachial flow-mediated dilation.*
- [23] A new clinical method for the assessment of penile endothelial function using the flow-mediated dilation with plethysmography technique. *J Urol*. *Human study in erectile dysfunction using venous occlusion strain gauge plethysmography of the penis with reactive hyperaemia after transient ischaemia.* [VERIFY: AUTHORS, YEAR, VOLUME AND PAGES NOT CONFIRMED; AVAILABILITY AND VALIDATION STATUS OF THE METHOD NOT ESTABLISHED]